

An analytical tool path smoothing algorithm for robotic machining with the consideration of redundant kinematics

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ARTICLE INFO

Keywords:

Robotic machining
Tool path
Kinematics
Redundancy
Spline

ABSTRACT

In the machining of complex parts with free-formed surfaces, robots are widely employed due to their advantages of a large operating space and high flexibility. The industrial robot with 6 degrees-of-freedom (DOF) has an extra redundant degree of freedom around the tool axis, which does not affect the tool pose related to the workpieces but influences the robot's joint configuration. The motion performance and machining efficiency can be improved by optimizing the redundant angle. Based on this, an analytical path smoothing algorithm for 6-DOF robots with the consideration of redundant kinematics is proposed to improve the robot's dynamic performance. The tool tip position and orientation are fit with the analytical 5th degree Pythagorean-Hodograph (PH) spline to satisfy C^2 continuity, respectively. Therefore, the 5th degree polynomial spline with minimum acceleration is adopted to fit the redundant rotation angle and the tool tip position arc length. Then the tool path position spline, tool orientation spline, and redundant rotary angle spline are all synchronized to the tool position displacement, which makes it convenient to do interpolation along the tool path. The minimum time feed planning method considering the joint dynamic constraints is adopted to interpolate motion commands. Experimental results show that the motion efficiency of the robot in the same test path increases by 33.97 % compared with the regular spline without considering acceleration of the redundant angle spline. Meanwhile, the proposed tool path smoothing method effectively mitigates the joint vibration with a maximum reduction of 65.05 %, without sacrificing motion accuracy.

1. Introduction

Traditional industrial robots with six degrees of freedom (DOF) are commonly used to transport workpieces from one position to another. The trajectory plan for such carrying motions usually requires precise control over velocity and acceleration, while maintaining precise contour accuracy during the motion process is less important. However, as industrial robots have improved in terms of positioning accuracy and structural rigidity, they are increasingly utilized in manufacturing industries including milling [1–4], polishing [5–9], and deburring [10, 11], which fully show their larger workspace, lower cost, and higher flexibility [12,13]. When it comes to 5-axis machining tasks, addressing the dynamics and motion precision throughout the entire tool path profile becomes critical [14,15]. The generation and optimization of the contour motion path in robotic machining present a significantly higher degree of complexity compared with the "point-to-point" type motions. In recent years, many researchers have made significant progress in this

field.

Generally, commercial software typically generates robot motion commands that contain discrete positions and orientations. However, tangential discontinuity in these discrete linear tool paths can result in fluctuating velocity and acceleration. This leads to significant vibrations in the robot's motion. Therefore, to ensure steady robotic machining motions, the tool path must be smoothed before being executed by the motor drives. Tool path smoothing methods can be categorized as either joint space methods or workpiece frame methods, depending on the operation space. In the joint space method, the robot motion commands of individual joints are smoothed directly [16–18]. This method allows for the direct execution of the smoothed tool path results by joint drives, eliminating the need for additional inverse kinematics computations. However, this method can make it challenging to restrain the deviation errors of the tool path profile when smoothing tool path commands of the joint due to the nonlinear kinematics. However, the workpiece frame method, involves direct smoothing of the tool path commands in the

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<https://doi.org/10.1016/j.rcim.2024.102768>

Received 27 December 2023; Received in revised form 21 March 2024; Accepted 25 March 2024

Available online 10 April 2024

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workpiece coordinate system (WCS) under the constraints of geometric deviation errors [19,20], and then the commands are transformed to joint displacements before being sent to joint drives [21–23]. This study focuses on the smoothing method of the 6-DOF robot in the WCS.

Unlike 3-axis machining, where smoothing is required only for the tool position commands, 5-axis or 6-axis machining tasks necessitate the smoothing of the tool orientation direction as well [24–26]. Based on the synchronization method of the tool path position and orientation spline, smoothing methods can be classified into two categories: coupled methods and decoupled methods. In the coupled method, the fitting of tool path position and orientation splines is accomplished using the same parameter. For example, Langeron et al. [27] developed dual B-splines to represent the tool pose, one spline was used to fit the tool path position, while another one was used to fit points equidistant to the tip point along the orientation vector. Wan et al. [28] proposed a corner smoothing algorithm to smooth the tool tip position and tool orientation by constructing asymmetrical PH splines, which resulted in higher machining efficiency. To ensure that the two splines have equal distances after fitting, Huang et al. [29] proposed a novel curve “airthoid” to smooth the corners of the tool position in the WCS. Bi et al. [30] introduced a B-spline in the quaternion space rather than the Cartesian space. However, the orientation may exhibit large fluctuations at short linear segments for complex surface parts. The coupled method cannot accurately reflect the simultaneous variance tendency, subsequently affecting the dynamic performance of the orientation, which has been proven by the study of Yuen et al. [31]. To address this issue, algorithms were developed to smooth the position and orientation individually, fully reflecting the fluctuation of the tool path position and orientation. Then the position and orientation spline parameters were mapped to the tool tip displacement to synchronize them along the tool path [32,33]. Li et al. [34] proposed the use of triple NURBS curves to fit the position and axis of the tool.

Unlike 5-axis machine tools, which are affected by only two rotary axes for tool-axis orientation, 6-DOF robots with their six rotary joints require more intricate processing. This is because the tool orientation may be affected by all six rotary joints, depending on the robot's kinematics. Yang et al. [35] introduced an analytical continuous tool path corner smoothing algorithm specifically designed for 6R robot manipulators. This algorithm involves the construction of micro B-splines to represent the tool path position in the Cartesian system, while the tool orientation is represented by B-splines in the Euler angle frame. To ensure synchronization between the tool-axis direction and tool tip position, the remaining linear segments are replaced with specially constructed B-splines. Using this approach, the algorithm achieves smooth tool path corners by coordinating the tool tip displacement with the tool axis direction and position. However, there are two issues need to be further solved in the study of tool path smoothing for robotic machining. First, compared with 5-axis CNC machining, robot machining has one redundant rotary freedom. Although the redundant rotary freedom does not change the tool pose in the WCS, it affects the robot's joint motions via inverse kinematics. Therefore, the smoothing of the redundant rotary angle directly affects the dynamic performance of the joint commands. While the discrete solution of the redundant rotary freedom has been studied [36,37], there are not tool path smoothing algorithms that consider the redundant rotation of 6R robot machining. Second, existing algorithms of 6R robot machining are mainly based on NURBS and B-splines, where iterative calculations are needed to estimate the tool length because the spline arc length lacks an explicit function with the spline parameter that restricts the calculation efficiency. Moreover, the estimation errors of tool lengths induce fluctuations of feed rate after tool path interpolation. Although some analytical tool path smoothing algorithms are reported to three-axis [38, 39] and five-axis machine tools [38] by constructing special splines that have analytical solutions of the arc-length, they cannot be directly adopted to 6-DOF robotic machining owing to the more complex and redundant kinematics of 6-DOF robot. Therefore, the development of

analytical robot path smoothing algorithms for robotic 5-axis machining task is crucial, especially when considering the presence of the redundant DOF in robotic machining. These algorithms play a significant role in achieving precise and efficient motion planning, ensuring smooth tool path execution, and optimizing overall machining performance in complex 5-axis machining tasks.

This paper proposes an analytical C^2 continuous path smoothing algorithm for robotic machining with the consideration of redundant kinematics. This paper makes several significant contributions to the field, which includes the following:

- (1) The proposed algorithm considers the redundant kinematics of a 6-DOF robot in order to optimize tool path smoothing for robotic 5-axis machining task.
- (2) The algorithm analytically calculates the tool path position displacements and control points of all curves, making it suitable for high efficiency calculations during robot machining processes.
- (3) The algorithm employs a 5th-degree polynomial spline with minimum acceleration to fit the redundant orientation angle to the arc length, ensuring lower vibrations in robotic machining.
- (4) The C^2 continuity of the tool path is achieved by optimizing the federate under the constraints of the joint velocity, acceleration and jerk, which results in acceleration continuity of the interpolated joint commands.

The remaining sections of this paper are organized as follows: [Section 2](#) provides the kinematics and redundancy analysis of robotic machining tasks. [Section 3](#) compares the characteristics of robotic machining with traditional 5-axis CNC machining, and details how the tool path position, orientation and redundant rotary commands are all smoothed analytically by constructing corresponding splines. [Section 4](#) presents experimental verifications. [Section 5](#) provides the conclusions.

2. Kinematics and redundancy of robotic machining

2.1. Kinematics analysis of robotic machining

As shown in [Fig. 1](#), in the multi-axis machining tasks, cutting location commands (CL data) consist of discrete tool path positions $P = [P_x, P_y, P_z]^T$ and orientations $O = [O_x, O_y, O_z]^T$ in the WCS generated by the commercial computer-aided-manufacture (CAM) software.

The transfer matrix of the cut location related to the workpiece frame

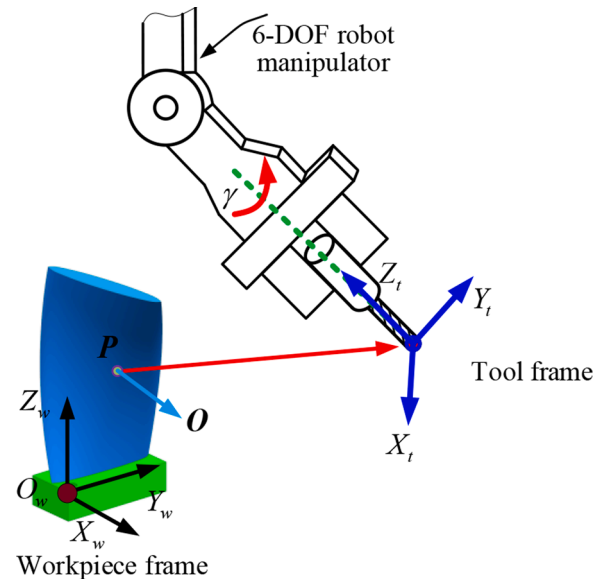


Fig. 1. Tool pose representation of robotic milling.

is expressed as:

$${}^w_p T(\theta) = \begin{bmatrix} R_{3 \times 3} & P \\ 0_{1 \times 3} & 1 \end{bmatrix} \quad (1)$$

where $R_{3 \times 3} = \begin{bmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{bmatrix}$ is the rotary transfer matrix of the tool frame related to the workpiece frame, and $P = [P_x, P_y, P_z]^T$ is the translation offset.

As shown in Fig. 2, the homogeneous transfer matrix of the tool frame related to the workpiece frame is expressed as:

$${}^w_p T(\theta) = [{}^w_p T]^{-1} \cdot {}^0_1 T(\theta_1) \cdot {}^1_2 T(\theta_2) \cdot {}^2_3 T(\theta_3) \cdot {}^3_4 T(\theta_4) \cdot {}^4_5 T(\theta_5) \cdot {}^5_6 T(\theta_6) \cdot {}^6_i T \quad (2)$$

In Eq. (2), ${}^0_{wp} T = \begin{bmatrix} I_{3 \times 3} & {}^0_{wp} P \\ 0_{1 \times 3} & 1 \end{bmatrix}$ is the transfer matrix of the workpiece

related to the base frame, ${}^6_i T = \begin{bmatrix} I_{1 \times 3} & {}^6_i P \\ 0_{1 \times 3} & 1 \end{bmatrix}$ is the transfer matrix of the tool frame related to the frame of the 6th axis. ${}^{i-1}_i T(\theta_i)$ ($i = 1, 2, \dots, 6$) is the transfer matrix of the i -th axis related to the $(i-1)$ -th axis, and θ_i is the rotary displacement of the i -th axis. ${}^{i-1}_i T(\theta_i)$ can be determined from the Denavit-Hartenberg (DH) method.

The robot task pose can be transferred to a vector $(P_x, P_y, P_z, \alpha, \beta, \gamma)$ with the Euler angle method, where (α, β, γ) are z - y - z Euler angles. The transfer matrix ${}^w_p T(\theta)$ is as follows:

$${}^w_p T(\theta) = Rot(z, \alpha) Rot(y, \beta) Rot(z, \gamma) Trans(P_x, P_y, P_z) \quad (3)$$

$$\text{where } Rot(z, \alpha) = \begin{bmatrix} C_\alpha & -S_\alpha & 0 & 0 \\ S_\alpha & C_\alpha & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, Rot(y, \beta) = \begin{bmatrix} C_\beta & 0 & S_\beta & 0 \\ 0 & 1 & 0 & 0 \\ -S_\beta & 0 & C_\beta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

represent the rotation transformation and $Trans(P_x, P_y, P_z) =$

$$\begin{bmatrix} 1 & 0 & 0 & P_x \\ 0 & 1 & 0 & P_y \\ 0 & 0 & 1 & P_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \text{ represents the translation transformation, } C_\alpha, S_\alpha, C_\beta,$$

$S_\beta, C_\gamma, S_\gamma$ are abbreviations of $\cos \alpha, \sin \alpha, \cos \beta, \sin \beta, \cos \gamma, \sin \gamma$ in this paper, respectively.

The result of Eq. (3) is:

$${}^w_p T(\theta) = \begin{bmatrix} C_\alpha C_\beta C_\gamma - S_\alpha S_\gamma & -C_\alpha C_\beta S_\gamma - S_\alpha C_\gamma & C_\alpha S_\beta & P_x \\ S_\alpha C_\beta C_\gamma + C_\alpha S_\gamma & -S_\alpha C_\beta S_\gamma + C_\alpha C_\gamma & S_\alpha S_\beta & P_y \\ -S_\beta C_\gamma & S_\beta S_\gamma & C_\beta & P_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (4)$$

Inverse kinematics model is needed to transfer the robot task pose to joint commands $\theta(t) = [\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6]^T$ in the robotic joint space by combining Eqs. (2), (3) and (4). The robotic machining process, as illustrated in Fig. 2, involves two kinematics chains—the robot chain and the workpiece chain. These two chains constitute the entire kinematics system and are transformed into each other via kinematic equations.

Combining Eq. (4), the tool orientation is calculated by using Euler angle as follows:

$$[O_x, O_y, O_z, 0]^T = {}^w_p T(\theta) [r_{ot}, 0]^T = [\cos \alpha \sin \beta, \sin \alpha \sin \beta, \cos \beta, 0]^T \quad (5)$$

where $r_{ot} = [0, 0, 1]$ is the tool orientation vector at the initial robot pose. For a given cutting location (CL) data $[P_x, P_y, P_z, O_x, O_y, O_z]^T$ generated by the CAM software, combining Eq. (5), the Euler angle α and β are calculated from elements of the vector:

$$\begin{cases} \alpha = \text{atan2}(O_y, O_x) \\ \beta = \text{atan2}(\sqrt{O_x^2 + O_y^2}, O_z) \end{cases} \quad (6)$$

From Eq. (4), it can be seen that the robot joint displacements θ are affected by the rotary freedom α, β , and γ via robot inverse kinematics. However, from Eqs. (5) and (6), the tool orientation vector is not directly affected by the Euler angle γ , which means that the rotational angle around the z -axis vector does not change the orientation vector of the tool related to the WCS (see Fig. 3). Therefore, although the rotary freedom γ affects the robot joint configuration, it is redundant to the tool orientation adjustment in robotic machining. By using the redundant kinematics character of rotation freedom, many researchers studied the redundant angle optimization to improve the machining performance by fully utilizing the pose-dependent character of robotic machining, for example, enlarging the structure stiffness [2,40], enhancing the tool path smoothness of the robotic machining [37], and reducing the deformation error in robotic milling [3].

In this paper, the initial redundant angles at discrete cutting locations are determined by using the tangential approximation method as

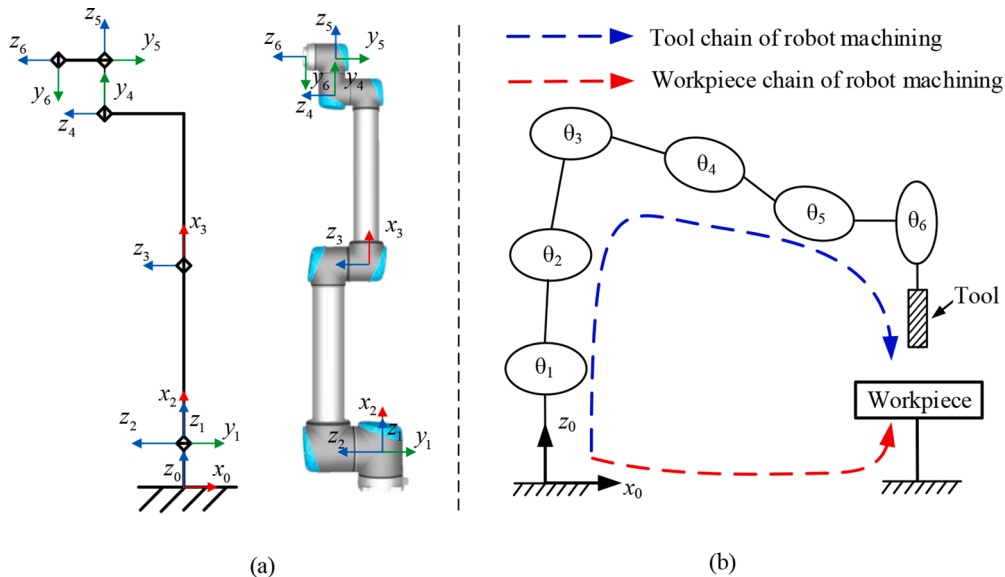


Fig. 2. Kinematic model: (a) Local coordinate system set on the robot; (b) Kinematic chains of robotic machining system.

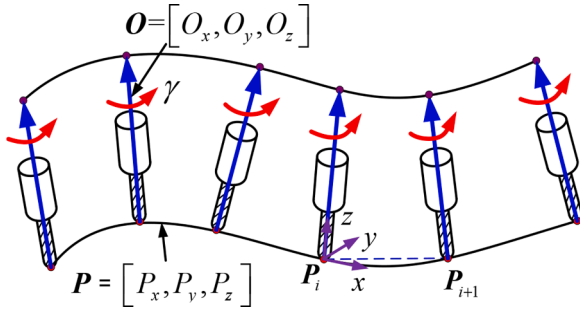


Fig. 3. Redundant rotary angle and cutting location data of robotic milling.

introduced in Ref. [37]. The specific procedures are as follows: as shown in Fig. 3, the tool tip position P_i is taken as the origin of the i th robot task pose; the tool orientation vector O_i is taken as the z -axis vector, i.e., $z_i = O_i$; the cross product of the tool orientation vector and the vector $P_i P_{i+1}$ is taken as the y -axis vector, i.e., $y_i = z_i \times (P_{i+1} - P_i) / \|z_i \times (P_{i+1} - P_i)\|$; and the x -axis vector is determined by the right hand rule $x_i = y_i \times z_i$. Finally, the robot task pose ${}^w T(\theta)$ is expressed as ${}^w T(\theta) = \begin{bmatrix} x_i & y_i & z_i & P_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$. Then, combining with Eq. (4), the initial Euler angle γ corresponding to the cutting location $[P_i, O_i]$ is calculated by $\gamma = \text{atan2}({}^w T_{32}, -{}^w T_{31})$.

Compared with the 5-axis task, the tool path of robotic machining with 6 DOF robot not only considers the position and orientation, the redundant rotary freedom γ (Fig. 3) also needs to be simultaneously considered to fully optimize the kinematic-dependent character of the robot machining system.

3. Tool path generation for 6-DOF robot machining

3.1. Overall introduction of the tool path smoothing

Due to the presence of the redundant rotary angle in the robotic machining tasks, this paper proposes a global analytical path smoothing method considering the redundant angle to fully harness the dynamic performance of the robot. The tool tip position is smoothed in Cartesian space by using a 5th degree Pythagorean-Hodograph (PH) spline $P(u)$ that is capable of online high-speed analytical calculation. To construct the tool orientation spline, the discrete orientation vector $O = [O_x, O_y, O_z]^T$ is firstly transformed into the Euler angle space $\Theta = [\alpha, \beta]^T$. Subsequently, the tool orientation is smoothed by creating a 5th degree plane PH spline $\Theta(\omega)$ in the Euler angle space. The redundant rotation angles are smoothed by constructing a 5th degree polynomial spline $\gamma(s)$ of the rotary angle γ related to the tool tip arc length s . Since both the position spline parameter u and redundant angle spline can establish analytical relationships with the tool tip arc length s , so only the tool orientation parameter ω is need to synchronize with s along the tool path. Therefore, the tool tip position, orientation and redundant rotation commands are smoothed and mapped to the tool tip displacements with functions $P(u(s))$, $\Theta(\omega(s))$ and $\gamma(s)$ respectively. By doing feed rate plan along the tool tip displacements, the time-stamped arc-length $s(t)$ is calculated. Finally, the time-stamped motion commands $\theta(t)$ of robot joints are obtained through robot kinematics based on the planed position commands $P(u(s(t)))$, orientation commands $\Theta(\omega(s(t)))$ and redundant angle commands $\gamma(s(t))$ are obtained. The flowchart of the proposed method is presented in Fig. 4.

3.2. C^2 continuous tool path position smoothing method

Because of the advantage to analytically calculate the tool path arc length using spline parameter, PH-spline is used to smooth the tool tip position. This approach ensures both the efficiency and accuracy of tool path interpolation.

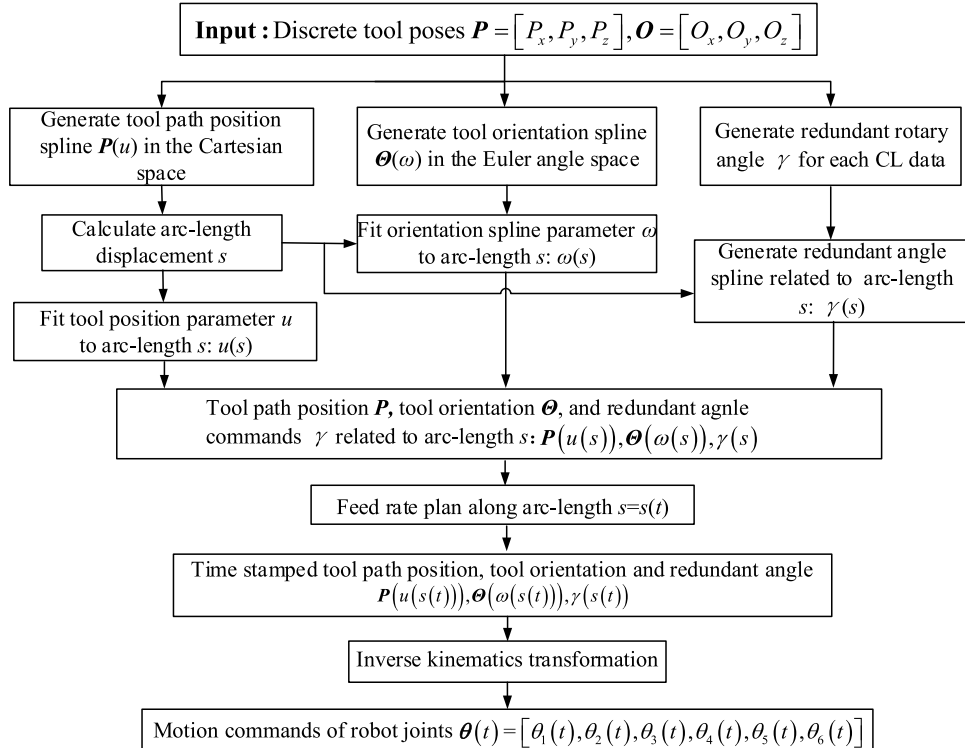


Fig. 4. Flowchart of the proposed robotic path smoothing algorithm with the consideration of redundant rotary angle motion.

A k -th degree PH-spline is defined as (for the i -th segment spline):

$$P_i(u) = \sum_{j=0}^k B_{i,j} b_j^k(u) \quad 0 \leq u \leq 1 \quad (7)$$

where u is the spline parameter, $\{B_{i,j}\}$ are the spline control points, and $b_j^k(u) = \frac{k!}{j!(k-j)!} u^j (1-u)^{k-j}$ are the k -th degree PH-spline basis functions, in this paper $k = 5$. The schematic of the PH-spline is shown in Fig. 5.

For more details about the PH-spline, the reader can refer to Ref. [41]. The 5th degree PH-spline curves are used to fit the tool tip position $P = [P_x, P_y, P_z]$. To construct the C^2 continuous PH spline for N segment splines, the quadratic polynomial $w_i(u)$ and quaternion coefficients $q_0, q_1, \dots, q_{N+1}$ are adopted to define the first derivative of $P_i(u)$:

$$P'_i(u) = w_i(u) i w_i^*(u) \quad (8)$$

The quaternion polynomial is expressed as follows:

$$w_i(u) = w_{i,0}(1-u)^2 + 2w_{i,1}(1-u)u + w_{i,2}u^2 \quad (9)$$

$$\begin{cases} w_{i,0} = \frac{1}{2}(q_{i-1} + q_i) \\ w_{i,1} = q_i \\ w_{i,2} = \frac{1}{2}(q_i + q_{i+1}) \end{cases}$$

The PH-spline is constructed by using the above method which can satisfy the C^2 continuity of the tool tip position. To calculate the quaternion coefficient q_i , the i -th segment PH spline is calculated by integrating the Eq. (8):

$$\begin{aligned} \int_0^1 P'_i(u) du &= \Delta P_i \\ &= \frac{1}{5} w_{i,0} i w_{i,0}^* + \frac{1}{10} (w_{i,0} i w_{i,1}^* + w_{i,1} i w_{i,0}^*) + \dots \\ &\quad \frac{1}{30} (w_{i,0} i w_{i,2}^* + 4w_{i,1} i w_{i,1}^* + w_{i,2} i w_{i,0}^*) + \dots \\ &\quad \frac{1}{10} (w_{i,1} i w_{i,2}^* + w_{i,2} i w_{i,1}^*) + \frac{1}{5} w_{i,2} i w_{i,2}^* \end{aligned} \quad (10)$$

Substituting Eq. (9) into Eq. (10):

$$6q_{i-1} i q_{i-1}^* + 13(q_{i-1} i q_i^* + q_i i q_{i-1}^*) + 54q_i i q_i^* + q_{i-1} i q_{i+1}^* + q_{i+1} i q_{i-1}^* + 13(q_i i q_{i+1}^* + q_{i+1} i q_i^*) + 6q_{i+1} i q_{i+1}^* = 120 \Delta P_i \quad (11)$$

For $i = 1, 2, \dots, N$

Assuming that the tangent vectors of the endpoints are t_0 and t_N , the expression for the initial and final boundary conditions are:

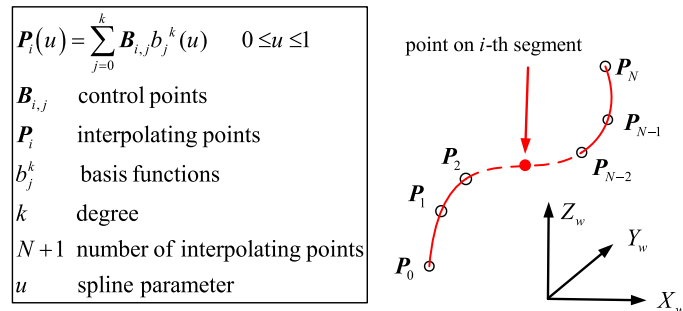


Fig. 5. Illustration of the PH-spline representation.

$$\begin{cases} P'_1(0) = \frac{1}{4}(q_0 + q_1) i (q_0 + q_1)^* = t_0 \\ P'_N(1) = \frac{1}{4}(q_N + q_{N+1}) i (q_N + q_{N+1})^* = t_N \end{cases} \quad (12)$$

Through Eq. (11) and Eq. (12), $N+2$ quaternion equations about $N+2$ unknown variables q are established. However, owing to the high order, these cannot directly be solved. Therefore, this paper adopts the Newton-like method in Ref [42] to solve q , the specific computation procedure is detailed in Appendix A. Substituting the q into Eq. (9), w is solved. The control points B are calculated by the following formulas:

$$\begin{cases} B_{i,0} = P_{i-1} \\ B_{i,1} = B_{i,0} + \frac{1}{5} w_{i,0} i w_{i,0}^* \\ B_{i,2} = B_{i,1} + \frac{1}{10} (w_{i,0} i w_{i,1}^* + w_{i,1} i w_{i,0}^*) \\ B_{i,3} = B_{i,2} + \frac{1}{30} (w_{i,0} i w_{i,2}^* + 4w_{i,1} i w_{i,1}^* + w_{i,2} i w_{i,0}^*) \\ B_{i,4} = B_{i,3} + \frac{1}{10} (w_{i,1} i w_{i,2}^* + w_{i,2} i w_{i,1}^*) \\ B_{i,5} = B_{i,4} + \frac{1}{5} w_{i,2} i w_{i,2}^* \end{cases} \quad (13)$$

Hence, a piecewise 5th degree PH spline curve that satisfies the C^2 continuity requirement is obtained. The points on the i -th PH spline are calculated as follows:

$$P_i(u) = \begin{bmatrix} x(u) \\ y(u) \\ z(u) \end{bmatrix} = \sum_{j=0}^k B_{i,j} b_j^k(u) \quad (14)$$

Meanwhile, to synchronize between arc-lengths and spline parameters, it is essential to calculate the arc length of the tool path position spline. One notable advantage of the PH spline lies in its straightforward analytical arc-length calculation [43], as demonstrated by the following equations:

$$\begin{cases} s_i(u) = \sum_{kk=0}^5 s_{i,kk} b_{kk}^5(u) \\ s_{i,0} = 0, s_{i,m} = \frac{1}{5} \sum_{j=0}^{m-1} \sigma_{i,j}, m = 1, \dots, 5 \end{cases} \quad (15)$$

where $\sigma_i = [\sigma_{i,0}, \sigma_{i,1}, \sigma_{i,2}, \sigma_{i,3}, \sigma_{i,4}]$ is the parametric speed, which is calculated by the following equations:

$$\begin{cases} \sigma_{i,0} = |w_{i,0}|^2 \\ \sigma_{i,1} = \frac{1}{2} (w_{i,0} i w_{i,1}^* + w_{i,1} i w_{i,0}^*) \\ \sigma_{i,2} = \frac{1}{6} (w_{i,0} i w_{i,2}^* + 4|w_{i,1}|^2 + w_{i,2} i w_{i,0}^*) \\ \sigma_{i,3} = \frac{1}{2} (w_{i,1} i w_{i,2}^* + w_{i,2} i w_{i,1}^*) \\ \sigma_{i,4} = |w_{i,2}|^2 \end{cases} \quad (16)$$

3.3. Smoothing of the tool orientation

The tool orientation $O_i = [O_{i,x}, O_{i,y}, O_{i,z}] (i = 0, 1, \dots, N)$ is a unit vector, and if directly fitted using spline interpolation, it cannot maintain the constraint of unit length, which leads to the loss of orientation accuracy. Therefore, the tool orientation cannot be directly interpolated in the Cartesian space. Therefore, to overcome this limitation, the tool orientation vectors are converted to the Euler rotary angle space $\Theta_i = [\alpha_i, \beta_i]$ by using Eq. (6).

The C^2 PH spline curves for the i th segment spline are constructed to

smooth the tool-axis orientation:

$$\Theta_i(\omega) = \begin{bmatrix} \alpha(\omega) \\ \beta(\omega) \end{bmatrix} = \sum_{j=0}^k \mathbf{M}_{i,j} b_j^k(\omega) \quad (17)$$

where ω is the spline parameter, $\{\mathbf{M}_{i,j}\}$ are the control points, and $b_j^k(\omega) = \frac{k!}{j!(k-j)!} \omega^j (1-\omega)^{k-j}$ are basis functions of the PH spline. Fig. 6 provides an illustration of the tool orientation spline representation.

To construct the PH spline, the quadratic polynomial $\mathbf{v}_i(\omega)$ is constructed with $N+2$ unknown complex variables $z_0, z_1, \dots, z_N, z_{N+1}$ to represent the derivatives of the PH spline:

$$\Theta_i'(\omega) = \mathbf{v}_i'(\omega) \quad (18)$$

where:

$$\mathbf{v}_i(\omega) = \mathbf{v}_{i,0}(1-\omega)^2 + 2\mathbf{v}_{i,1}(1-\omega)\omega + \mathbf{v}_{i,2}\omega^2$$

$$\begin{cases} \mathbf{v}_{i,0} = \frac{1}{2}(\mathbf{z}_{i-1} + \mathbf{z}_i) \\ \mathbf{v}_{i,1} = \mathbf{z}_i \\ \mathbf{v}_{i,2} = \frac{1}{2}(\mathbf{z}_i + \mathbf{z}_{i+1}) \end{cases} \quad (19)$$

The PH-spline constructed using the above method can satisfy the C^2 continuity requirement of the i th tool-axis orientation segment spline. Thereafter, to calculate the coefficients $\mathbf{z}$, integrate the Eq. (18):

$$\int_0^1 \Theta_i'(\omega) d\omega = \Delta \Theta_i = \begin{bmatrix} \alpha_i - \alpha_{i-1} \\ \beta_i - \beta_{i-1} \end{bmatrix} \quad (20)$$

$$= \frac{1}{20}\mathbf{z}_{i-1}^2 + \frac{9}{20}\mathbf{z}_i^2 + \frac{1}{20}\mathbf{z}_{i+1}^2 + \frac{1}{60}\mathbf{z}_{i-1}\mathbf{z}_{i+1} + \frac{13}{60}\mathbf{z}_{i-1}\mathbf{z}_i + \frac{13}{60}\mathbf{z}_i\mathbf{z}_{i+1}$$

where $i = 1, \dots, N$.

Assuming that the tangent vectors of the endpoints are $\mathbf{t}_0$ and $\mathbf{t}_N$, the expression for the initial and final boundary conditions are:

$$\begin{cases} \Theta_1'(0) = \frac{1}{4}(\mathbf{z}_0 + \mathbf{z}_1)^2 = \mathbf{t}_0 \\ \Theta_N'(1) = \frac{1}{4}(\mathbf{z}_N + \mathbf{z}_{N+1})^2 = \mathbf{t}_N \end{cases} \quad (21)$$

Combining Eqs. (20) and (21), $N+2$ quaternion equations about $N+2$ unknown variables $\mathbf{z}$ are established. Since the derivative of the Eqs. (20), (21) about $\mathbf{z}$ is directly obtained analytically, the solution of $\mathbf{z} = (z_0, \dots, z_{N+1})$ is calculated directly by using Newton-Raphson method in Ref. [44]. Substituting $\mathbf{z}$ into Eq. (19), $\mathbf{v}_i(\omega)$ is calculated, and the control points $\mathbf{M}$ are obtained by:

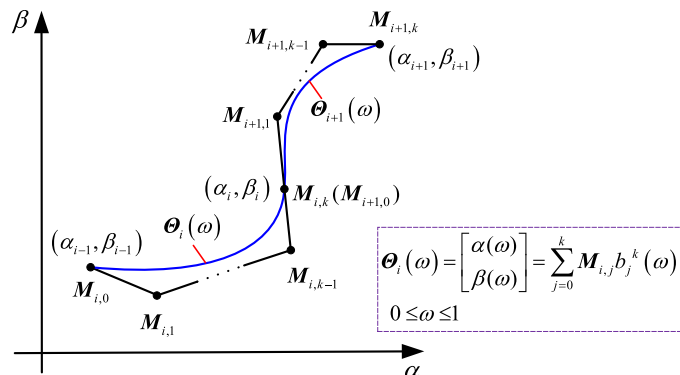


Fig. 6. Illustration of PH-spline representation for tool-axis orientation.

$$\begin{cases} \mathbf{M}_{i,0} = \Theta_{i-1} \\ \mathbf{M}_{i,1} = \mathbf{M}_{i,0} + \frac{1}{5}\mathbf{v}_{i,0}^2 \\ \mathbf{M}_{i,2} = \mathbf{M}_{i,1} + \frac{1}{5}\mathbf{v}_{i,0}\mathbf{v}_{i,1} \\ \mathbf{M}_{i,3} = \mathbf{M}_{i,2} + \frac{1}{15}(\mathbf{v}_{i,0}\mathbf{v}_{i,2} + 2\mathbf{v}_{i,1}^2) \\ \mathbf{M}_{i,4} = \mathbf{M}_{i,3} + \frac{1}{5}\mathbf{v}_{i,2}\mathbf{v}_{i,1} \\ \mathbf{M}_{i,5} = \mathbf{M}_{i,4} + \frac{1}{5}\mathbf{v}_{i,2}^2 \end{cases} \quad (22)$$

By substituting the control points $\mathbf{M}$ into Eq. (17), the PH spline $\Theta_i(\omega)$ for the tool orientation in the Euler angle space is obtained.

3.4. Smoothing of the redundant angle

Due to the single-variable nature of the redundant angle γ , it is synchronized with the arc-length parameter s by employing a 5th degree piecewise polynomial fitting approach to satisfy $\gamma_i(s_i) = \gamma_i(i = 1, 2, \dots, N)$. By adjusting the coefficients of the polynomial, the C^2 continuity and minimum acceleration is obtained, thereby ensuring high dynamics along the tool path. The schematic diagram is depicted in Fig. 7, where $\gamma_i(s) = \sum_{j=0}^5 c_{i,j}s^{5-j}$ is a 5th degree polynomial in the i -th segment $[0, s_i]$, ($i = 1, 2, \dots, N$).

To achieve the good dynamic performance of the redundant angle, the global index of the second order differential of the redundant angle related to the arclength Φ is minimized as follows:

$$\Phi = \sum_{i=1}^N \int_0^{s_i} \left| \frac{d^2\gamma}{ds^2} \right|^2 ds \quad (23)$$

The 1st and 2nd derivatives of the redundant angle spline are expressed as follows:

$$\begin{cases} \gamma_i'(s) = [5s^4 \ 4s^3 \ 3s^2 \ 2s \ 1 \ 0] \cdot \mathbf{C}_i \\ \gamma_i''(s) = [20s^3 \ 12s^2 \ 6s \ 2 \ 0 \ 0] \cdot \mathbf{C}_i \end{cases} \quad (24)$$

where $\mathbf{C}_i = [c_{i,0}, c_{i,1}, c_{i,2}, c_{i,3}, c_{i,4}, c_{i,5}]^T$ are the polynomial coefficients.

By combining Eqs. (23) and (24) of the i th segment spline is expressed as follows:

$$\Phi_i = \int_0^{s_i} \left\{ \begin{aligned} &400c_{i,0}^2s^6 + 480c_{i,0}c_{i,1}s^5 + 240c_{i,0}c_{i,2}s^4 + \dots \\ &144c_{i,1}^2s^4 + (80c_{i,0}c_{i,3} + 144c_{i,1}c_{i,2})s^3 + \dots \\ &(48c_{i,1}c_{i,3} + 36c_{i,2}^2)s^2 + 24c_{i,2}c_{i,3}s + 4c_{i,3}^2 \end{aligned} \right\} ds \quad (25)$$

Eq. (25) is transformed into the form of matrix multiplication:

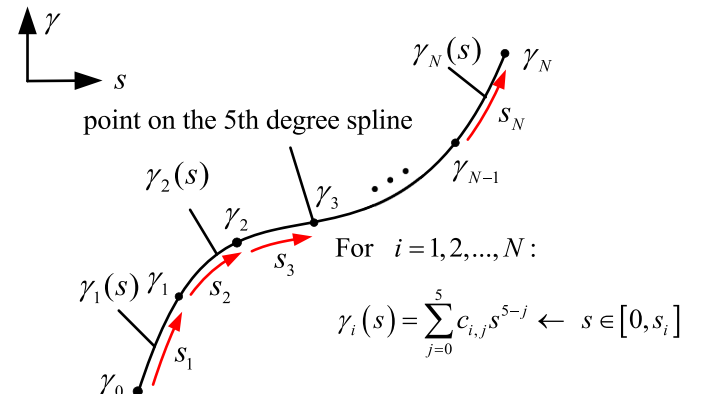


Fig. 7. Fitting of redundant angle related to the arc length.

$$\Phi_i = \begin{bmatrix} c_{i,0} \\ c_{i,1} \\ c_{i,2} \\ c_{i,3} \\ c_{i,4} \\ c_{i,5} \end{bmatrix}^T \underbrace{\begin{bmatrix} \frac{400}{7}s_i^7 & 40s_i^6 & 24s_i^5 & 10s_i^4 & 0 & 0 \\ 40s_i^6 & \frac{144}{5}s_i^5 & 18s_i^4 & 8s_i^3 & 0 & 0 \\ 24s_i^5 & 18s_i^4 & 12s_i^3 & 6s_i^2 & 0 & 0 \\ 10s_i^4 & 8s_i^3 & 6s_i^2 & 4s_i & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}}_{Q_i} \begin{bmatrix} c_{i,0} \\ c_{i,1} \\ c_{i,2} \\ c_{i,3} \\ c_{i,4} \\ c_{i,5} \end{bmatrix} \quad (26)$$

Since this optimization function is a quadratic minimization (QP) program, in order to simplify the operation, Eq. (26) is written as:

$$\Phi_i = \frac{1}{2} C_i^T Q_i C_i \quad (27)$$

Therefore, the global optimization function is expressed as:

$$\min \Phi = \min \frac{1}{2} C^T Q C \quad (28)$$

The spline of $\gamma(s)$ should satisfy C^2 continuity, therefore, the corresponding linear constraints need to be added. Firstly, position constraints are added to ensure that the spline passes through the discrete point (s_i, γ_i) :

$$\gamma_i(0) = \gamma_{i-1}, \gamma_i(s_i) = \gamma_i \text{ For } i = 1, 2, \dots, N-1, N \quad (29)$$

The matrix form representation of Eq. (29) can be expressed as:

$$\underbrace{\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ s^5 & s^4 & s^3 & s^2 & s & 0 \end{bmatrix}}_{L_i^0} = C_i = \underbrace{\begin{bmatrix} 0 \\ s_i \end{bmatrix}}_{\tau_i^0} \text{ For } i = 1, 2, \dots, N-1, N \quad (30)$$

Therefore, the expressions of the global position constraints are as follows:

$$\underbrace{\begin{bmatrix} L_1^0 & & 0 \\ & L_2^0 & \\ & & \ddots \\ 0 & & & L_N^0 \end{bmatrix}}_{L^0} C = \underbrace{\begin{bmatrix} \tau_1^0 \\ \tau_2^0 \\ \vdots \\ \tau_N^0 \end{bmatrix}}_{\tau^0} \quad (31)$$

In order to satisfy the velocity and acceleration continuity constraints of the adjacent polynomial splines, the following constraints should be satisfied:

$$\begin{cases} \gamma'_i(0) = \gamma'_{i-1}(s_{i-1}) \\ \gamma''_i(0) = \gamma''_{i-1}(s_{i-1}) \end{cases} \text{ For } i = 2, 3, \dots, N-1 \quad (32)$$

Similarly, the constraints in Eq. (32) are rewritten into the matrix form:

$$\underbrace{\begin{bmatrix} 5s_{i-1}^4 & 4s_{i-1}^3 & 3s_{i-1}^2 & 2s_{i-1} & 1 & 0 & : & 0 & 0 & 0 & 0 & -1 & 0 \\ 20s_{i-1}^3 & 12s_{i-1}^2 & 6s_{i-1} & 2 & 0 & 0 & : & 0 & 0 & 0 & -2 & 0 & 0 \end{bmatrix}}_{L_{i-1}^{12}} \begin{bmatrix} C_{i-1} \\ C_i \end{bmatrix} = \underbrace{\begin{bmatrix} 0 \\ 0 \end{bmatrix}}_{\tau_{i-1}^{12}} \quad (33)$$

Therefore, based on Eq. (33), the global constraints of the velocity and acceleration continuity are written as:

$$\underbrace{\begin{bmatrix} L_1^{12} & & 0 \\ & L_2^{12} & \\ & & \ddots \\ 0 & & & L_{N-1}^{12} \end{bmatrix}}_{L^{12}} C = \underbrace{\begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}}_{\tau^{12}} \quad (34)$$

The first and second derivatives at the initial and final endpoints are set to 0:

$$\underbrace{\begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & : & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & : & 0 & 0 & 0 & 0 & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & : & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & 0 & 0 & 0 & : & 5s_N^4 & 4s_N^3 & 3s_N^2 & 2s_N & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & : & 20s_N^3 & 12s_N^2 & 6s_N & 2 & 0 & 0 \end{bmatrix}}_{L^{end}} \begin{bmatrix} C_1 \\ C_1 \\ \vdots \\ C_N \\ C_N \end{bmatrix} = \underbrace{\begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix}}_{\tau^{end}} \quad (35)$$

Combining with Eqs. (31), (34) and (35), the optimization problem is written as follows:

$$\begin{aligned} \min \Phi &= \min \frac{1}{2} C^T Q C \\ \text{s.t. } \begin{bmatrix} L^0 \\ L^{12} \\ L^{end} \end{bmatrix} C &= \begin{bmatrix} \tau^0 \\ \tau^{12} \\ \tau^{end} \end{bmatrix} \end{aligned} \quad (36)$$

The above quadratic minimization problem can be solved with the MATLAB Optimization Toolbox by using the fmincon function.

The three types of splines established using the above method consist three different spline parameters, in order to facilitate the subsequent feed velocity planning, it is necessary to establish the three spline parameters (u, ω, γ) as functions of the arc length s . Since the redundant angle spline $\gamma(s)$ is already a function of s , only $u(s)$ and $\omega(s)$ need to be constructed. First, for $u(s)$, because the constructed PH spline has solution of the arc length s related to the spline parameter u , (see Eq. (14)). The Newton-Raphson method is used to invert Eq. (15) to get $u(s)$. The arc length range in segment i is defined as s_i , therefore, the solution of u_t corresponding to s_t is converted to find the root of the equation $ss(u) = 0$:

$$ss(u) = s(u) - s_t \quad (37)$$

where s_t is the given arc length that needs to calculate the corresponding spline parameter u_t .

The iterative equation is:

$$u_{kk} = u_{kk-1} - \frac{ss(u_{kk-1})}{ss'(u_{kk-1})} \quad (38)$$

where kk is the iteration number, $ss'(u) = s'(u)$, (see Eq. (15)).

The tool-axis orientation does not directly have an analytical expression with the arc length, so the polynomial fitting method in [31] is used here just to establish the function $\omega(s)$:

$$\omega_i(s) = \sum_{j=0}^5 d_{ij} s^{5-j} \quad (39)$$

where $d_i = [d_{i,0}, d_{i,1}, \dots, d_{i,5}]$ denotes the polynomial coefficients of the i

th segment.

Therefore, the tool path position, orientation and redundant rotation angle commands are all smoothed and mapped to the tool tip displacements with functions $P(u(s))$, $\Theta(\omega(s))$ and $\gamma(s)$ as shown in Fig. 8.

3.5. Interpolation with drive constraints of industrial robot

In the Sections 3.1~3.4, the tool position, orientation, and redundant angle splines are fitted to the arc-length s smoothly. The next step is to implement feed planning along the arc length s . In this study, a global feed planning algorithm is employed to realize feed planning which is achieved by the adoption of a minimum time feed optimization method considering joint dynamic constraints. This method aims to minimize the robot motion time along the trajectory while respecting the joint displacement, velocity, acceleration and jerk constraints imposed by the robotic joint movements, thereby achieving efficient and smooth motion commands:

$$\min \int_0^s \frac{ds}{s} \quad s.t. \begin{cases} \theta_{\min} \leq \theta(s) \leq \theta_{\max} \\ |\dot{\theta}(s)| \leq v_{\max} \\ |\ddot{\theta}(s)| \leq a_{\max} \\ |\dddot{\theta}(s)| \leq j_{\max} \end{cases} \quad (40)$$

where $\theta_{\min}$, $\theta_{\max}$, $v_{\max}$, $a_{\max}$, $j_{\max}$ denote the joint displacement, velocity, acceleration and jerk constraints. $\theta(s)$, $\dot{\theta}(s)$, $\ddot{\theta}(s)$ denote the joint displacement, velocity, acceleration and jerk related to the arc length s which are calculated by:

$$\begin{cases} \dot{\theta}(s) = \frac{d\theta(s)}{ds} s' \\ \ddot{\theta}(s) = \frac{d^2\theta(s)}{ds^2} (s')^2 + \frac{d\theta(s)}{ds} s'' \\ \dddot{\theta}(s) = \frac{d^3\theta(s)}{ds^3} (s')^3 + 3 \frac{d^2\theta(s)}{ds^2} s' s'' + \frac{d\theta(s)}{ds} s''' \end{cases} \quad (41)$$

where $s' = \frac{ds}{dt}$, $s'' = \frac{d^2s}{dt^2}$, $s''' = \frac{d^3s}{dt^3}$ denote the first, second and third differentials of the arc length s related to time t .

The above optimization function in Eq. (40) is a nonlinear optimization function which is solved using Sequential Quadratic Problem (SQP) in Ref. [45]. The output of velocity planning is the robot end pose $[P(u(s(t))), \Theta(\omega(s(t))), \gamma(s(t))]$ with time stamp t , which is subsequently used to compute robot joint commands $\theta(t)$ through robot inverse

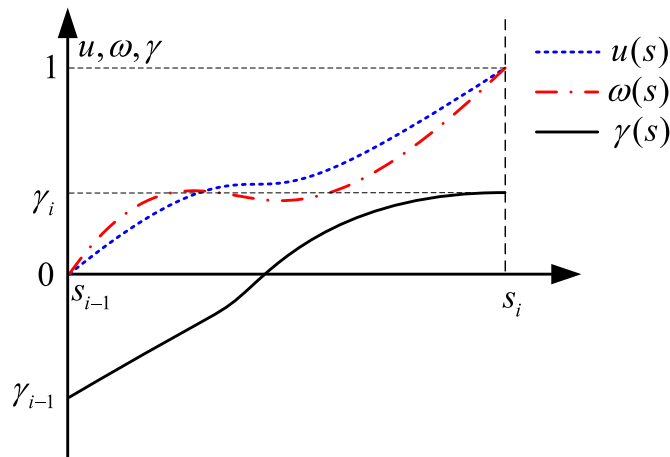


Fig. 8. Parameter synchronization function of the tool tip position $u(s)$, tool orientation $\omega(s)$ and redundant angle $\gamma(s)$.

kinematics. The complete procedure of the proposed tool path generation and smoothing method is shown in Fig. 4.

4. Experiments

The proposed algorithm is implemented in MATLAB, version 2020, on a Windows 11 system. The experiment is conducted on a UR10 robot with a 3 kg load as shown in Fig. 9. The interpolation period in the feed planning is precisely set to be 0.008 s. The constraints of the robot position are set as $v_{\max} = 25\text{mm/s}$, $a_{\max} = 2000\text{mm/s}^2$, $j_{\max} = 10000\text{mm/s}^3$. The joint constraints are listed in Table 1.

The test path is shown in Fig. 10, while Fig. 11 presents the global smoothed tool position spline, orientation spline, and redundant angle spline. To demonstrate the high efficiency calculation capability of the global PH spline, the total time required for the spline construction and the arc length calculation is compared with that of the B-spline method. It takes 74.8 ms for the proposed algorithm to finish the calculation, which is significantly faster than the B-spline method that requires 240.1 ms. The comparative results indicate that the proposed method can achieve highly accurate online calculations with improved efficiency.

To demonstrate the benefits of the proposed redundant angle smoothing algorithm with the consideration of acceleration constraints, C^2 continuous spline for (s, γ) without the consideration of the acceleration constraints of the redundant angle is compared here.

The same feed planning method is applied to interpolate the joint motion commands of the robot while adhering to the corresponding constraints of tangential velocity, acceleration, and jerk. The result of the feed planning is presented in Fig. 12, which shows that the average velocity achieved by the proposed algorithm is 14.337 mm/s. This is faster than the regular spline method without the consideration of the acceleration constraints of the redundant angle that achieves an average velocity of 9.467 mm/s. As a consequence of this increase in velocity, the motion cycle of the proposed method is reduced by 33.97%, from 26.728 s to 17.648 s, when compared with the regular C^2 continuous spline method. These results indicate that the proposed method effectively guarantees the motion dynamic constraints of the robot and enhances motion efficiency.

Practical path experiments are conducted on the actual robot using the test path, as shown in Fig. 10. The real-time joint positions and velocities of the robot are collected using the Python function package “urrtde package”. The actual joint acceleration shown in Fig. 13 is calculated by differentiating the velocity of each joint. The maximum acceleration of each joint is drawn with a bar chart as shown in Fig. 14. It can be seen that the maximum acceleration of other joints decreased significantly except the joint 4 and joint 5 which is approximately at the

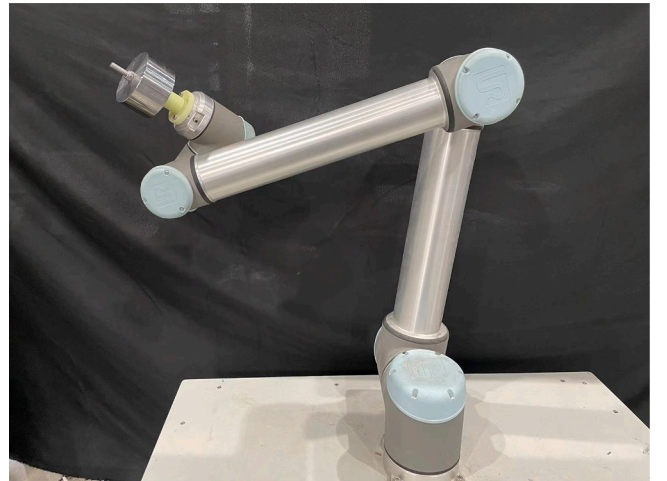


Fig. 9. Experimental platform used in the tests.

Table 1
Joint kinematic constraints of the experimental UR 10 robot.

Joint	1	2	3	4	5	6
Position [°]	[-360,360]	[-360,360]	[-360,360]	[-360,360]	[-360,360]	[-360,360]
Velocity [°/s]	120	120	120	180	180	180
Acceleration [°/s ²]	2000	2000	2000	2000	2000	2000
Jerk [°/s ³]	10,000	10,000	10,000	10,000	10,000	10,000

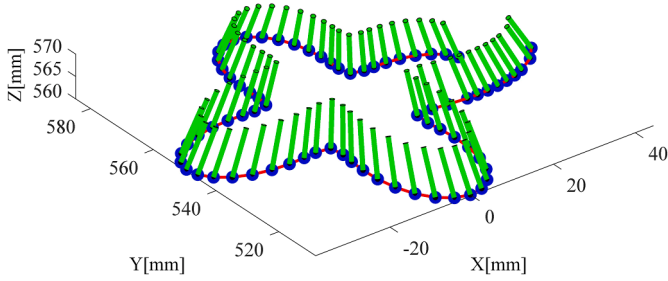


Fig. 10. The test tool path.

same level between regular spline and the proposed method. The results demonstrate that the method proposed in this study not only enhances the efficiency of robotic motion during tasks, but also effectively mitigates the joint acceleration with a maximum reduction of 65.05%, thereby reducing joint vibrations and enhancing the smoothness of joint movements.

To further validate that the proposed method does not compromise the motion accuracy of the robot while enhancing its efficiency, the target joint values θ_{target} and actual joint values θ_{actual} are collected

during the experiment by reading motor encoders during the movement with the "ur-rtde package" interface provided by the Universal Robots, and the joint tracking error e_θ of the robot is calculated as the difference between the reference and actual joint positions along the test path by:

$$e_\theta = \theta_{target} - \theta_{actual} \quad (42)$$

As shown in Fig. 15, Fig. 16, it is evident that the method proposed in this study outperforms the traditional spline in terms of joint tracking error peak values. This result demonstrates that by optimizing the acceleration of redundant angle splines, the proposed algorithm effectively enhances the robot's motion efficiency and reduce the joint vibrations without sacrificing motion accuracy.

Because the joint tracking error may not directly reflect the motion accuracy of the end effector, to further validate that the proposed method does not compromise the motion accuracy of the end effector while enhancing its efficiency, the tracking error of the end effector is calculated as the difference between the target position and the actual position of the end effector, while the contour error of the end effector is calculated as the closest distance between the target position and the actual position of the end effector with the algorithm in Ref. [46]. The target position and the actual position are calculated by using the target joint values θ_{target} and actual joint values θ_{actual} based on the forward

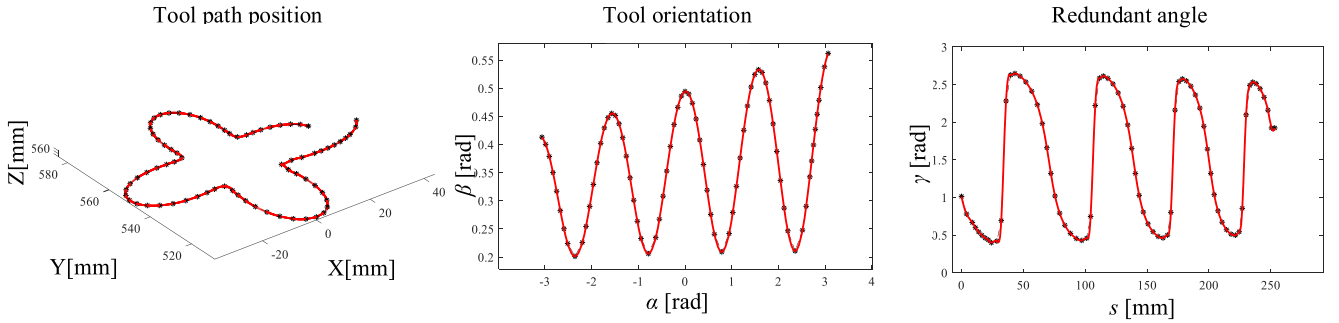


Fig. 11. The smoothed tool path position, orientation and redundant angle.

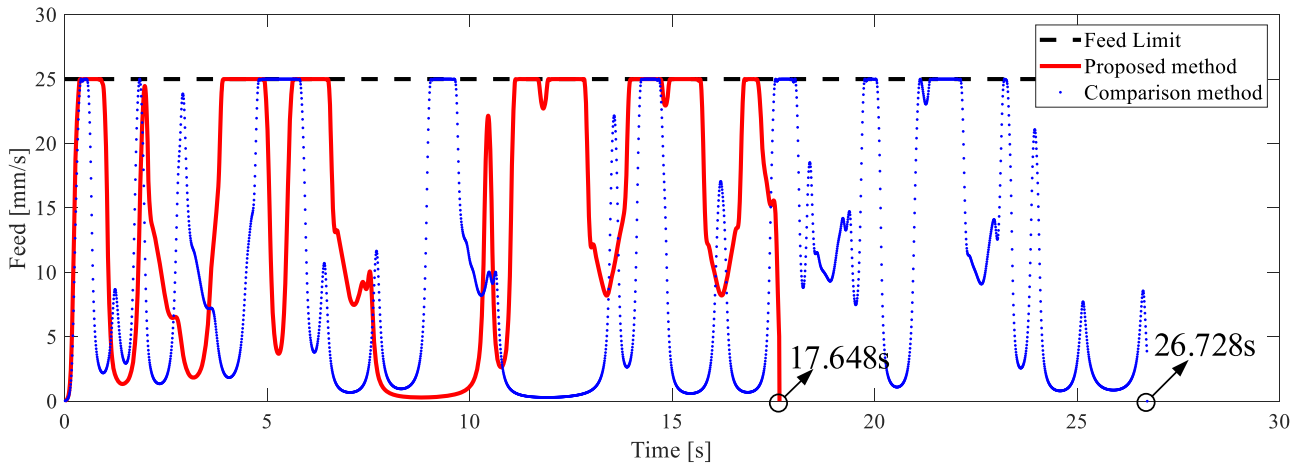


Fig. 12. Tangential velocities of the proposed method and the comparison method.

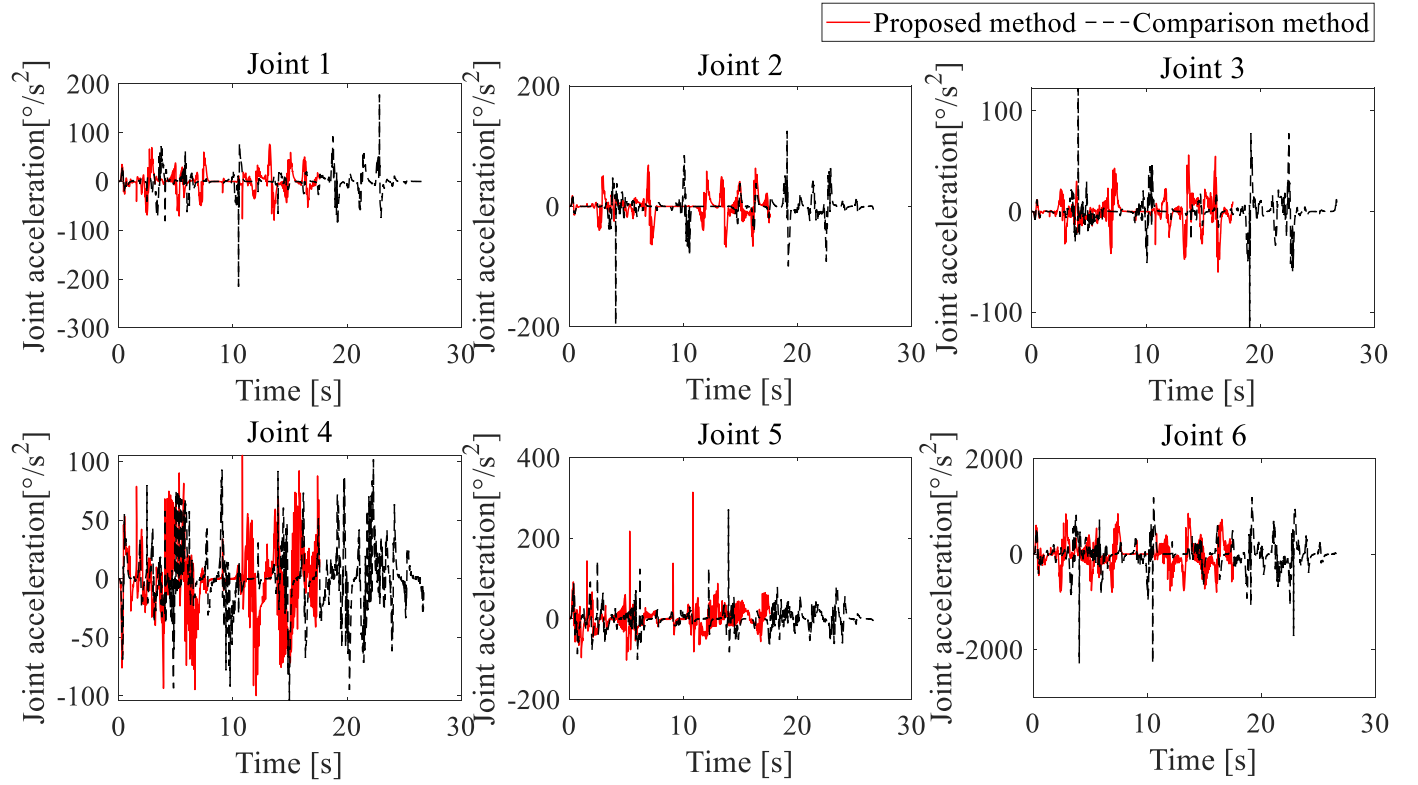


Fig. 13. Acceleration of each joint along the test path.

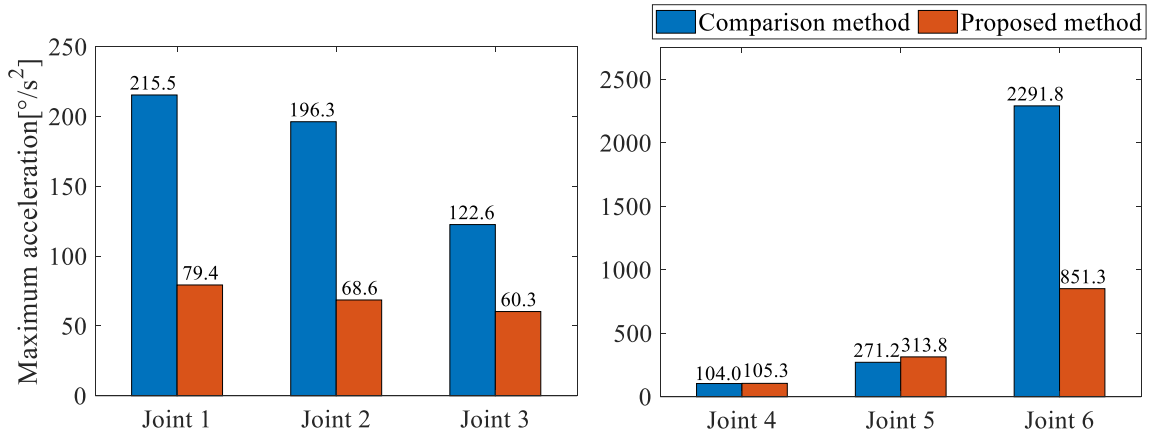


Fig. 14. Maximum acceleration of each joint along the test path.

kinematics transformation in Eq. (2). The tracking error and contour error of the end effector are as shown in Fig. 17. The results show that the contour error and tracking error of the end effector are at the same level between the proposed method and the comparison method, which proves that the proposed algorithm improves the efficiency without sacrificing the motion accuracy of the end effector.

5. Conclusion

This article presents an analytical C^2 continuous tool path smoothing algorithm for robotic machining with the consideration of redundant kinematics. A 5th degree PH spline is used to smooth the tool path position and orientation that satisfies C^2 continuity. To achieve smooth and continuous motion, the redundant rotation angles are smoothed using a 5th degree polynomial spline as a function of the tool position

displacement. This polynomial spline is optimized to minimize global acceleration. Synchronizing the tool path position spline parameter, the tool orientation spline parameter, and the redundant angle spline parameter along the path are mapped to the tool tip position displacements. The global minimum time feed planning method considering the joint velocity, acceleration, and jerk constraints is adopted to interpolate motion commands. The algorithm effectively reduces abrupt motion changes and ensures a smoother movement of the robot. The proposed method shows a 33.97% higher motion efficiency compared to the regular PH spline method. Furthermore, experiments demonstrate that the proposed method effectively mitigates the joint vibration with a maximum reduction of 65.05%, without compromising motion accuracy.

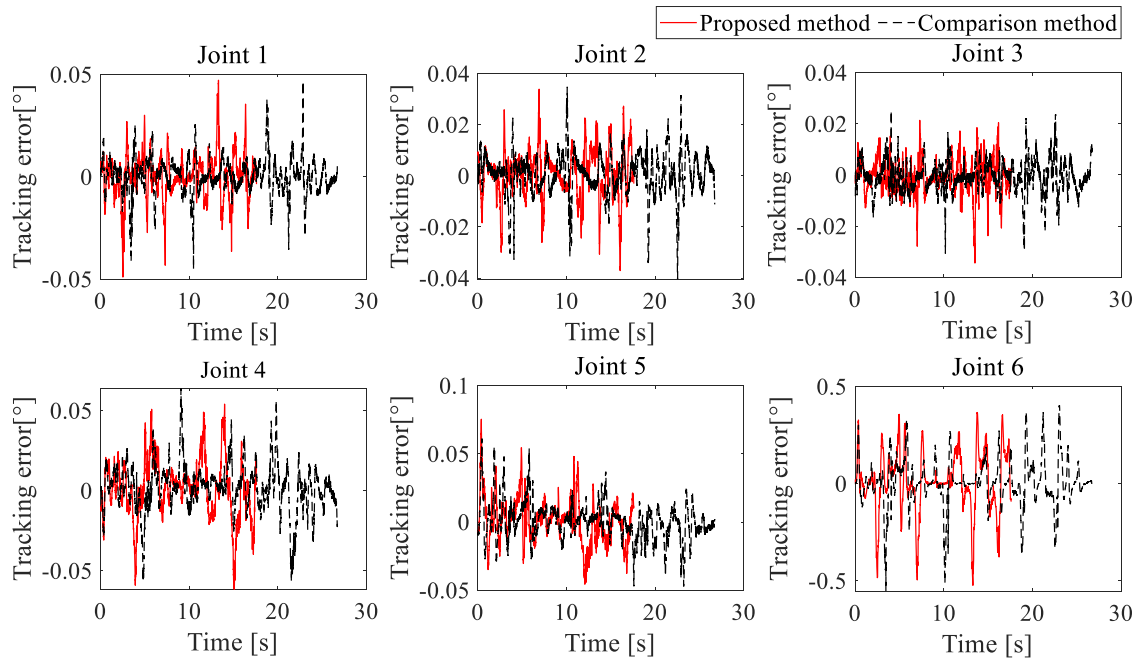


Fig. 15. Tracking error of each joint along the test path.

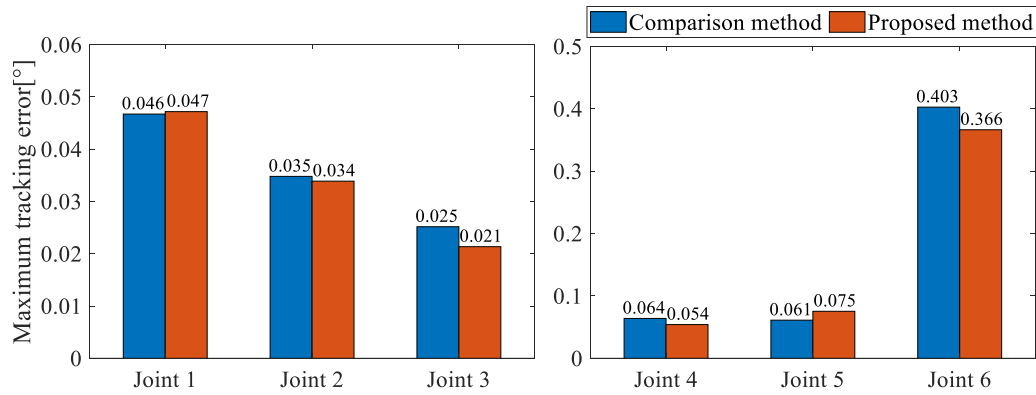


Fig. 16. Maximum tracking error of each joint along the test path.

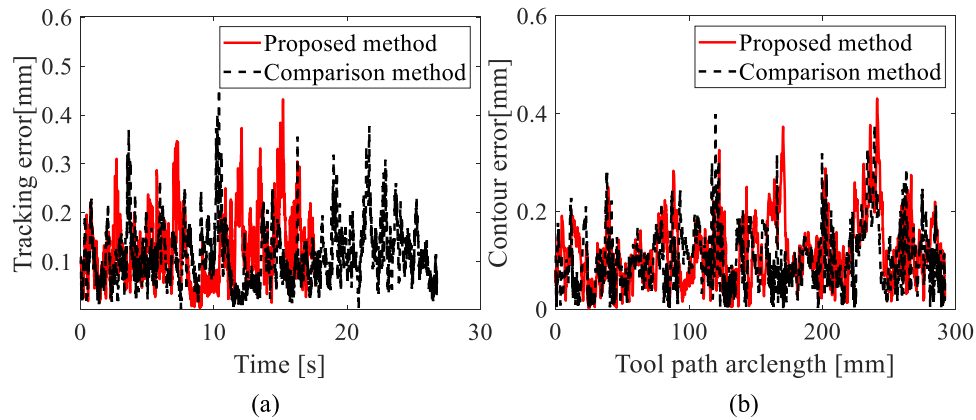


Fig. 17. Error of the end effector: (a) Tracking error of the end effector; (b) Contour error of the end effector.

CRediT authorship contribution statement

Jixiang Yang: Project administration, Methodology, Conceptualization, Investigation, Resources, Supervision, Writing – review & editing. **Qi Qi:** Data curation, Methodology, Validation, Writing – original draft. **Abulikemu Adili:** Data curation, Methodology, Validation, Writing – original draft. **Han Ding:** Methodology, Project administration, Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no conflicts of interest to this work.

We declare that we do not have any commercial or associative

interest that represents a conflict of interest in connection with the work submitted.

Data availability

Data will be made available on request.

Acknowledgement

This research is supported by the National Natural Science Foundation of China under Grant Nos. 52188102, 52122512 and 52375496, and the Natural Science Foundation of Hubei Province, China, under Grant No. 2021CFA075.

Appendix A. Newton-like method for calculating q

Step 1: Selection of the initial solution

Due to the similarity of the shape between a regular cubic B-spline curve and a 5th degree Pythagorean-Hodograph (PH) spline curve, the tool path position is often smoothed using a regular cubic B-spline curve as an initial solution to the PH-spline curve. Firstly, constructing the piecewise cubic B-spline $c_i(u)$, $c'(u = 0, 0.5, 1)$ is calculated by using the geometric properties of the B-spline curve. Then the initial conditions according to Eq. (8) are given by the following equations:

$$\begin{cases} P'_1(0) = \frac{1}{4}(q_0 + q_1)i(q_0 + q_1)^* = c'_1(0) \\ P'_i(0.5) = \frac{1}{64}(q_{i-1} + 6q_i + q_{i+1})i(q_{i-1} + 6q_i + q_{i+1})^* = c'_i(0.5) \\ P'_N(1) = \frac{1}{4}(q_N + q_{N+1})i(q_N + q_{N+1})^* = c'_N(1) \end{cases} \quad (A.1)$$

where $i = 1, 2, \dots, N$

The solution of the quaternion equations in Eq (A.1) can be treated as the calculation task of $AiA^* = c$. The solution is calculated as [47]:

$$A = \sqrt{\frac{1}{2}(1 + \lambda)|c|} \left(-\sin\varphi + \cos\varphi i - \frac{u\cos\varphi + v\sin\varphi}{1 + \lambda}j + \frac{v\cos\varphi - u\sin\varphi}{1 + \lambda}k \right) \quad (A.2)$$

where $(\lambda, u, v) = c/|c|$, $-\pi \leq \varphi \leq \pi$. $\varphi = -\frac{\pi}{2}$ is chosen here to obtain the initial solutions which are calculated as:

$$A = \sqrt{\frac{1}{2}(1 + \lambda)|c|} \left(1 - \frac{v}{1 + \lambda}j + \frac{u}{1 + \lambda}k \right) \quad (A.3)$$

Substituting Eq. (A.3) into Eq. (A.1), by solving the linear equations, the initial solution $q^{kk=0}$ can be obtained, where kk is the iteration number.

Step 2: Construction of the iteration equations and calculation

After obtaining the initial solution q^0 for iteration, the next step is to construct the iteration equations. The iteration equations are formed by combining Eqs. (11) and (12):

$$\begin{cases} P'_1(0) = \frac{1}{4}(q_0 + q_1)i(q_0 + q_1)^* = t_0 \\ 6q_{i-1}i q_{i-1}^* + 13(q_{i-1}i q_i^* + q_i i q_{i-1}^*) + 54q_i i q_i^* + (q_{i-1}i q_{i+1}^* + q_{i+1}i q_{i-1}^*) + \dots \\ 13(q_i i q_{i+1}^* + q_{i+1}i q_i^*) + 6q_{i+1}i q_{i+1}^* = 120\Delta P_i \\ P'_N(1) = \frac{1}{4}(q_N + q_{N+1})i(q_N + q_{N+1})^* = t_N \end{cases} \quad (A.4)$$

where $i = 1, 2, \dots, N$

Because the iterative equation in Eq. (A.4) contains the complex number i that cannot be directly differentiated. The linearization method is adopted to realize the iterative calculation. Assuming that q_i^{kk} is the quaternion coefficient after the iteration calculation of the kk -th step, here is the following linearized approximation equation:

$$\begin{aligned} q_i^{kk} i q_i^{kk*} &= (q_i^{kk} - q_i^{kk-1}) i q_i^{kk-1*} + q_i^{kk-1} i (q_i^{kk} - q_i^{kk-1})^* + q_i^{kk-1} i q_i^{kk-1*} + (q_i^{kk} - q_i^{kk-1}) i (q_i^{kk} - q_i^{kk-1})^* \\ &\approx (q_i^{kk} - q_i^{kk-1}) i q_i^{kk-1*} + q_i^{kk-1} i (q_i^{kk} - q_i^{kk-1})^* + q_i^{kk-1} i q_i^{kk-1*} \\ q_i^{kk} i q_j^{kk*} + q_j^{kk} i q_i^{kk*} &\approx 2q_j^{kk-1} i (q_i^{kk} - q_i^{kk-1})^* + 2q_i^{kk-1} i (q_j^{kk} - q_j^{kk-1})^* + q_i^{kk-1} i q_j^{kk-1*} + q_j^{kk-1} i q_i^{kk-1*} \end{aligned} \quad (A.5)$$

By substituting the above linearized equations into Eq. (A.4), $N+2$ linear equations are established. The iterative equations about $q_{i-1}^{kk}, q_i^{kk}, q_{i+1}^{kk}$ ($i = 1, 2, \dots, N$) can be obtained:

$$\begin{cases} \frac{1}{4}(\mathbf{q}_0^{kk} + \mathbf{q}_1^{kk})\mathbf{i}(\mathbf{q}_0^{kk} + \mathbf{q}_1^{kk})^* = \mathbf{c}'_1(0) \\ \mathbf{q}_0^{kk}\mathbf{i}Ea_i + \mathbf{q}_1^{kk}\mathbf{i}Eb_i + \mathbf{q}_1^{kk}\mathbf{i}Ec_i = Q_1 \\ \vdots \\ \mathbf{q}_{i-1}^{kk}\mathbf{i}Ea_i + \mathbf{q}_i^{kk}\mathbf{i}Eb_i + \mathbf{q}_{i+1}^{kk}\mathbf{i}Ec_i = Q_i \\ \vdots \\ \mathbf{q}_{N-1}^{kk}\mathbf{i}Ea_i + \mathbf{q}_N^{kk}\mathbf{i}Eb_i + \mathbf{q}_{N+1}^{kk}\mathbf{i}Ec_i = Q_N \\ \frac{1}{4}(\mathbf{q}_N^{kk} + \mathbf{q}_{N+1}^{kk})\mathbf{i}(\mathbf{q}_N^{kk} + \mathbf{q}_{N+1}^{kk})^* = \mathbf{c}'_N(1) \end{cases} \quad (\text{A.6})$$

where:

$$\begin{cases} Ea_i = (6\mathbf{q}_{i-1}^{kk-1} + 6\mathbf{q}_{i-1}^{kk-1*} + 26\mathbf{q}_i^{kk-1} + 2\mathbf{q}_{i+1}^{kk-1})^* \\ Eb_i = (26\mathbf{q}_{i-1}^{kk-1} + 54\mathbf{q}_i^{kk-1} + 54\mathbf{q}_i^{kk-1*} + 26\mathbf{q}_{i+1}^{kk-1})^* \\ Ec_i = (6\mathbf{q}_{i-1}^{kk-1} + 6\mathbf{q}_{i-1}^{kk-1*} + 26\mathbf{q}_i^{kk-1} + 2\mathbf{q}_{i+1}^{kk-1})^* \\ Q_i = 120\Delta P_i + 6\mathbf{q}_{i-1}^{kk-1}\mathbf{i}\mathbf{q}_{i-1}^{kk-1*} + 26\mathbf{q}_{i-1}^{kk-1}\mathbf{i}\mathbf{q}_i^{kk-1*} + 54\mathbf{q}_i^{kk-1}\mathbf{i}\mathbf{q}_i^{kk-1*} + 26\mathbf{q}_i^{kk-1}\mathbf{i}\mathbf{q}_{i+1}^{kk-1*} + \dots \\ 2\mathbf{q}_{i-1}^{kk-1}\mathbf{i}\mathbf{q}_{i+1}^{kk-1*} + 6\mathbf{q}_{i+1}^{kk-1}\mathbf{i}\mathbf{q}_{i+1}^{kk-1*} \end{cases} \quad (\text{A.7})$$

The kk -th iteration accuracy is calculated as:

$$\zeta_{kk} = \sqrt{\sum_{i=1}^{N+1} \|\mathbf{q}_i^{kk} - \mathbf{q}_i^{kk-1}\|^2} \quad (\text{A.8})$$

According to the preset iteration precision $\xi_{\min}$ and the maximum number of iterations $kk_{\max}$, the quaternion coefficients $\mathbf{q}$ are obtained through iterative calculations.

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